



11225 - A MIRI search for TWA 7b twins: sub-Jupiter mass planets shaping debris disks

Cycle: 5, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
TWA10				
	1	BKG-TWA10	MIRI Coronagraphic Imaging	(9) BKG-TWA10
	2	TWA10	MIRI Coronagraphic Imaging	(1) TWA10
	3	TWA10_PSF_reference	MIRI Coronagraphic Imaging	(2) TWA10_psf_reference
	4	BKG-TWA10_PSF_reference	MIRI Coronagraphic Imaging	(10) BKG-TWA10_psf_reference

JWST Proposal 11225 (Created: Monday, June 15, 2026, 1:00:28PM Eastern Standard Time) - Overview

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
2MJ0227				
	5	BKG-2MJ0227	MIRI Coronagraphic Imaging	(11) BKG-2MJ0227
	6	2MJ0227	MIRI Coronagraphic Imaging	(3) 2MJ0227
	7	2MJ0227_PSF_reference	MIRI Coronagraphic Imaging	(4) 2MJ0227_psf_reference
	8	BKG-2MJ0227_PSF_reference	MIRI Coronagraphic Imaging	(12) BKG-2MJ0227_psf_reference
Fomalhaut C				
	9	BKG_Fomalhaut_C	MIRI Coronagraphic Imaging	(13) BKG-Fomalhaut_C
	10	Fomalhaut_C	MIRI Coronagraphic Imaging	(5) Fomalhaut_C
	11	Fomalhaut_C_PSF_reference	MIRI Coronagraphic Imaging	(6) Fomalhaut_C_psf_reference
	12	BKG-Fomalhaut_PSF_reference	MIRI Coronagraphic Imaging	(14) BKG-Fomalhaut_C_psf_reference
AU Mic				
	13	BKG-AU_Mic	MIRI Coronagraphic Imaging	(15) BKG-AU_Mic
	14	AU_Mic	MIRI Coronagraphic Imaging	(7) AU_Mic
	15	AU_Mic_PSF_reference	MIRI Coronagraphic Imaging	(8) AU_Mic_psf_reference
	16	BKG-AU_Mic_PSF_reference	MIRI Coronagraphic Imaging	(16) BKG-AU_Mic_psf_reference

ABSTRACT

We will leverage the unparalleled mid-infrared coronagraphic sensitivity of MIRI to detect sub-Jupiter mass exoplanets in four close, young M-dwarf stars that host circumstellar debris disks. The recent discovery of TWA 7b, a 0.3 Jupiter mass exoplanet, and the first exoplanet discovery with JWST, highlights that young M-dwarfs with circumstellar disks are the ideal sample to directly image extrasolar planets at mid-infrared wavelengths. Our sample comprises the last four young M dwarfs with debris disks that have not yet been characterized by MIRI, and is unique because: 1) the lower intrinsic luminosity of these M-dwarf hosts, combined with the 10-15 micron sensitivity of MIRI, means that our observations will be sensitive to the atmospheres of cooler ice-giant analogues to our own solar system planets; 2) all of our targets have well-characterized circumstellar debris disks verifying that these targets are sites of efficient planetesimal formation, providing a testing ground to dynamically infer the mass of these planets, ultimately leading to powerful characterization.

OBSERVING DESCRIPTION

We will use the F1140C MIRI coronagraph which offers unprecedented sensitivity and contrast at mid-infrared wavelengths making possible the detection of sub-Jupiter mass planets orbiting stars in close proximity to the Sun. Our on-sky contrast estimates indicate that we will be easily sensitive to sub-Jupiter mass exoplanets for all four targets. We are assuming one hour (3600s) of observation time on each science target, allowing us to reach the deep contrasts necessary to be sensitive to this population of sub-Jupiter planets.

PSF Reference stars: We have selected a unique individual reference star for each of our science targets that are closely matched in spectral type, are relatively close on the sky, and have several indications that they are not multiple systems.

Nine-point dithers: For each of our reference star observations, we will use a nine-point dither pattern which gives significantly greater PSF diversity capturing the varying morphology of the residual scattered starlight depending on small offsets of the star behind the coronagraph. We have also verified that our reference stars are 1-2 magnitudes brighter than our science targets in the WISE W3 bands, ensuring that a nine-point dither pattern will not be prohibitively time expensive.

Glowstick: To mitigate the well-known ``glowstic’’ effect seen in all MIRI images, we have adopted the (now standard) practice of acquiring background observations immediately before the science target and immediately after the observations of the PSF reference star. All of our observations are linked in a non-interruptible sequence.

Proposal 11225 - Targets - A MIRI search for TWA 7b twins: sub-Jupiter mass planets shaping debris disks

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	TWA10	RA: 12 35 4.2560 (188.7677333d) Dec: -41 36 38.62 (-41.61073d) Equinox: J2000	Proper Motion RA: -69.524 mas/yr Proper Motion Dec: -29.560999996647297 mas/yr Parallax: 0.0173964" Epoch of Position: 2000	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[Circumstellar disks, Circumstellar dust, Debris disks, M stars] Extended=NO				
	(2)	TWA10_psf_reference	RA: 12 44 14.3343 (191.0597262d) Dec: -37 14 13.60 (-37.23711d) Equinox: J2000	Proper Motion RA: -90.753 mas/yr Proper Motion Dec: -42.23799990086263 mas/yr Parallax: 0.0094099" Epoch of Position: 2000	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[K dwarfs] Extended=NO				
	(3)	2MJ0227	RA: 02 27 29.2543 (36.8718929d) Dec: +30 58 24.60 (30.97350d) Equinox: J2000	Proper Motion RA: 79.542 mas/yr Proper Motion Dec: -72.12999994408165 mas/yr Parallax: 0.024423200000000003" Epoch of Position: 2000	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[Circumstellar disks, Circumstellar dust, Debris disks, K stars, M dwarfs]				
	(4)	2MJ0227_psf_reference	RA: 02 02 18.8576 (30.5785733d) Dec: +37 21 39.39 (37.36094d) Equinox: J2000	Proper Motion RA: -4.974 mas/yr Proper Motion Dec: -6.6029999516104 mas/yr Parallax: 7.354000000000001E-4" Epoch of Position: 2000	
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[K dwarfs, K stars] Extended=NO					

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(5)	Fomalhaut_C	RA: 22 48 4.4929 (342.0187204d) Dec: -24 22 7.72 (-24.36881d) Equinox: J2000	Proper Motion RA: 331.609 mas/yr Proper Motion Dec: -183.80500009698153 mas/yr Parallax: 0.13027070000000002" Epoch of Position: 2000
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[Circumstellar dust, Circumstellar gas, Debris disks, M dwarfs] Extended=NO			
(6)	Fomalhaut_C_psf_reference	RA: 23 32 49.3982 (353.2058258d) Dec: -16 50 44.32 (-16.84564d) Equinox: J2000	Proper Motion RA: 341.147 mas/yr Proper Motion Dec: -219.10799994202534 mas/yr Parallax: 0.06873560000000001" Epoch of Position: 2000
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[K dwarfs] Extended=NO			
(7)	AU_Mic	RA: 20 45 9.5325 (311.2897188d) Dec: -31 20 27.24 (-31.34090d) Equinox: J2000	Proper Motion RA: 281.319 mas/yr Proper Motion Dec: -360.1480000497759 mas/yr Parallax: 0.1029432" Epoch of Position: 2000
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[Circumstellar disks, Circumstellar dust, Debris disks, M dwarfs] Extended=NO			
(8)	AU_Mic_psf_reference	RA: 20 13 53.3962 (303.4724842d) Dec: -45 09 50.47 (-45.16402d) Equinox: J2000	Proper Motion RA: 778.331 mas/yr Proper Motion Dec: -159.93899990007776 mas/yr Parallax: 0.16221709999999998" Epoch of Position: 2000
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[M dwarfs, M stars] Extended=NO			
(9)	BKG-TWA10	RA: 12 34 57.0000 (188.7375000d) Dec: -41 35 0.00 (-41.58333d) Equinox: J2000	
Comments: Category=Calibration Description=[Coronagraphic, Telescope/sky background] Extended=NO			

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(10)	BKG-TWA10_psf_reference	RA: 12 44 30.0000 (191.1250000d) Dec: -37 12 17.00 (-37.20472d) Equinox: J2000
Comments: Category=Calibration Description=[Coronagraphic, Telescope/sky background] Extended=NO		
(11)	BKG-2MJ0227	RA: 02 27 18.0000 (36.8250000d) Dec: +30 54 27.00 (30.90750d) Equinox: J2000
Comments: Category=Calibration Description=[Coronagraphic, Telescope/sky background] Extended=NO		
(12)	BKG-2MJ0227_psf_reference	RA: 02 02 56.0000 (30.7333333d) Dec: +37 23 35.00 (37.39306d) Equinox: J2000
Comments: Category=Calibration Description=[Coronagraphic, Telescope/sky background] Extended=NO		
(13)	BKG-Fomalhaut_C	RA: 22 48 42.0000 (342.1750000d) Dec: -24 22 52.00 (-24.38111d) Equinox: J2000
Comments: Category=Calibration Description=[Coronagraphic, Telescope/sky background] Extended=NO		
(14)	BKG-Fomalhaut_C_psf_reference	RA: 23 33 13.0000 (353.3041667d) Dec: -16 57 5.00 (-16.95139d) Equinox: J2000
Comments: Category=Calibration Description=[Coronagraphic, Telescope/sky background] Extended=NO		
(15)	BKG-AU_Mic	RA: 20 44 36.0000 (311.1500000d) Dec: -31 25 5.00 (-31.41806d) Equinox: J2000
Comments: Category=Calibration Description=[Coronagraphic, Telescope/sky background] Extended=NO		
(16)	BKG-AU_Mic_psf_reference	RA: 20 13 26.1000 (303.3587500d) Dec: -45 08 38.15 (-45.14393d) Equinox: J2000
Comments: Category=Calibration Description=[Coronagraphic, Telescope/sky background] Extended=NO		

Proposal 11225 - Observation 1 - A MIRI search for TWA 7b twins: sub-Jupiter mass planets shaping debris disks

Observation	Proposal 11225, Observation 1: BKG-TWA10												Mon Jun 15 18:00:28 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Coronagraphic Imaging												
	Background Observation For: [TWA10 (Obs 2)]												
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous		
	(9)	BKG-TWA10	RA: 12 34 57.0000 (188.7375000d)										
			Dec: -41 35 0.00 (-41.58333d)										
			Equinox: J2000										
	Comments: Category=Calibration Description=[Coronagraphic, Telescope/sky background] Extended=NO												
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Repeat observation					Background Quadrant						
		YES					1						
Dithers	#	Dither Type											
	1	BACKGROUND											
Spectral Elements	#	Coron Mask/Filter	Subarray	Mask	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	4QPM/F1140C	MASK1140	4QPM	F1140C	FASTR1	1251	12	1	2	24	7201.425	276758.2
PSF References	Additional Justification: false												

Proposal 11225 - Observation 1 - A MIRI search for TWA 7b twins: sub-Jupiter mass planets shaping debris disks

Special Requirements	No Parallel Attachments Sequence Observations 1, 2, 3, 4, Non-interruptible
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Proposal 11225 - Observation 2 - A MIRI search for TWA 7b twins: sub-Jupiter mass planets shaping debris disks

Observation	Proposal 11225, Observation 2: TWA10												Mon Jun 15 18:00:28 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Coronagraphic Imaging												
	Background Observations:[BKG-TWA10 (Obs 1)]												
Diagnostics	(TWA10 (Obs 2)) Warning (Form): Science observations should be linked to at least one other compatible science observation by an Aperture PA Offset of 1-14 degrees												
	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous			
	(1)	TWA10	RA: 12 35 4.2560 (188.7677333d) Dec: -41 36 38.62 (-41.61073d) Equinox: J2000				Proper Motion RA: -69.524 mas/yr Proper Motion Dec: -29.560999996647297 mas/yr Parallax: 0.0173964" Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.												
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.												
	Category=Star Description=[Circumstellar disks, Circumstellar dust, Debris disks, M stars] Extended=NO												
Acquisition	#	Target	Filter	Quadrant	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID			
	1	1 TWA10	FND	1	FAST	4	1	1	0.959	276578.1			
Template	Repeat observation												
	NO												
Dithers	#	Dither Type											
	1	NONE											
Spectral Elements	#	Coron Mask/Filter	Subarray	Mask	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	4QPM/F1140C	MASK1140	4QPM	F1140C	FASTR1	1251	12	1	1	12	3600.713	276578.2

Proposal 11225 - Observation 2 - A MIRI search for TWA 7b twins: sub-Jupiter mass planets shaping debris disks

PSF References	TWA10_PSF_reference (Obs 3) (PSF Reference; Filters [F1140C]) Additional Justification: false
Special Requirements	Aperture PA Range 80.45 to 100.45 Degrees (V3 75.61455103 to 95.61455103) No Parallel Attachments Sequence Observations 1, 2, 3, 4, Non-interruptible

Proposal 11225 - Observation 3 - A MIRI search for TWA 7b twins: sub-Jupiter mass planets shaping debris disks

Observation	Proposal 11225, Observation 3: TWA10_PSF_reference											Mon Jun 15 18:00:28 GMT 2026	
	Diagnostic Status: Warning												
	Observing Template: MIRI Coronagraphic Imaging												
	Background Observations:[BKG-TWA10_PSF_reference (Obs 4)]												
Diagnostics	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous			
	(2)	TWA10_psf_reference	RA: 12 44 14.3343 (191.0597262d) Dec: -37 14 13.60 (-37.23711d) Equinox: J2000				Proper Motion RA: -90.753 mas/yr Proper Motion Dec: -42.23799990086263 mas/yr Parallax: 0.0094099" Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.												
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.												
	Category=Star Description=[K dwarfs] Extended=NO												
Acquisition	#	Target	Filter	Quadrant	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID			
	1	2 TWA10_psf_refere nce	FND	1	FAST	4	1	1	0.959	276758.3			
Template	Repeat observation												
	NO												
Dithers	#	Dither Type											
	1	9-POINT-SMALL-GRID											
Spectral Elements	#	Coron Mask/Filter	Subarray	Mask	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	4QPM/F1140C	MASK1140	4QPM	F1140C	FASTR1	832	5	1	9	45	8982.248	276758.4

Proposal 11225 - Observation 3 - A MIRI search for TWA 7b twins: sub-Jupiter mass planets shaping debris disks

PSF References	PSF Reference: true
Special Requirements	No Parallel Attachments Sequence Observations 1, 2, 3, 4, Non-interruptible

Proposal 11225 - Observation 4 - A MIRI search for TWA 7b twins: sub-Jupiter mass planets shaping debris disks

Observation	Proposal 11225, Observation 4: BKG-TWA10_PSF_reference												Mon Jun 15 18:00:28 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Coronagraphic Imaging												
	Background Observation For: [TWA10_PSF_reference (Obs 3)]												
Diagnostics	(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates					Targ. Coord. Corrections			Miscellaneous		
	(10)	BKG-TWA10_psf_reference	RA: 12 44 30.0000 (191.1250000d) Dec: -37 12 17.00 (-37.20472d) Equinox: J2000										
	Comments: Category=Calibration Description=[Coronagraphic, Telescope/sky background] Extended=NO												
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter		Repeat observation					Background Quadrant					
	F1000W		YES					1					
Dithers	#	Dither Type											
	1	BACKGROUND											
Spectral Elements	#	Coron Mask/Filter	Subarray	Mask	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	4QPM/F1140C	MASK1140	4QPM	F1140C	FASTR1	832	5	1	2	10	1996.055	276758.4
PSF References	Additional Justification: false												

Proposal 11225 - Observation 4 - A MIRI search for TWA 7b twins: sub-Jupiter mass planets shaping debris disks

Special Requirements	No Parallel Attachments Sequence Observations 1, 2, 3, 4, Non-interruptible
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Proposal 11225 - Observation 5 - A MIRI search for TWA 7b twins: sub-Jupiter mass planets shaping debris disks

Observation	Proposal 11225, Observation 5: BKG-2MJ0227 Mon Jun 15 18:00:28 GMT 2026												
	Diagnostic Status: Warning Observing Template: MIRI Coronagraphic Imaging Background Observation For: [2MJ0227 (Obs 6)]												
Diagnostics	(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous			
	(11)	BKG-2MJ0227	RA: 02 27 18.0000 (36.8250000d) Dec: +30 54 27.00 (30.90750d) Equinox: J2000 <i>Comments:</i> <i>Category=Calibration</i> <i>Description=[Coronagraphic, Telescope/sky background]</i> <i>Extended=NO</i>										
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Repeat observation				Background Quadrant							
		YES				1							
Dithers	#	Dither Type											
	1	BACKGROUND											
Spectral Elements	#	Coron Mask/Filter	Subarray	Mask	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	4QPM/F1140C	MASK1140	4QPM	F1140C	FASTR1	1251	12	1	2	24	7201.425	276758.6
PSF References	Additional Justification: false												

Proposal 11225 - Observation 5 - A MIRI search for TWA 7b twins: sub-Jupiter mass planets shaping debris disks

Special Requirements	No Parallel Attachments Sequence Observations 5, 6, 7, 8, Non-interruptible
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Proposal 11225 - Observation 6 - A MIRI search for TWA 7b twins: sub-Jupiter mass planets shaping debris disks

Observation	Proposal 11225, Observation 6: 2MJ0227												Mon Jun 15 18:00:28 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Coronagraphic Imaging												
	Background Observations:[BKG-2MJ0227 (Obs 5)]												
Diagnostics	(2MJ0227 (Obs 6)) Warning (Form): Science observations should be linked to at least one other compatible science observation by an Aperture PA Offset of 1-14 degrees												
	(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous			
	(3)	2MJ0227	RA: 02 27 29.2543 (36.8718929d) Dec: +30 58 24.60 (30.97350d) Equinox: J2000				Proper Motion RA: 79.542 mas/yr Proper Motion Dec: -72.12999994408165 mas/yr Parallax: 0.024423200000000003" Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.												
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.												
	Category=Star Description=[Circumstellar disks, Circumstellar dust, Debris disks, K stars, M dwarfs]												
Acquisition	#	Target	Filter	Quadrant	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID			
	1	3 2MJ0227	FND	1	FAST	4	1	1	0.959	276758.5			
Template	Repeat observation												
	NO												
Dithers	#	Dither Type											
	1	NONE											
Spectral Elements	#	Coron Mask/Filter	Subarray	Mask	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	4QPM/F1140C	MASK1140	4QPM	F1140C	FASTR1	1251	12	1	1	12	3600.713	276758.6

Proposal 11225 - Observation 6 - A MIRI search for TWA 7b twins: sub-Jupiter mass planets shaping debris disks

PSF References	2MJ0227_PSF_reference (Obs 7) (PSF Reference; Filters [F1140C]) Additional Justification: false
Special Requirements	No Parallel Attachments Sequence Observations 5, 6, 7, 8, Non-interruptible

Proposal 11225 - Observation 7 - A MIRI search for TWA 7b twins: sub-Jupiter mass planets shaping debris disks

Observation	Proposal 11225, Observation 7: 2MJ0227_PSF_reference											Mon Jun 15 18:00:28 GMT 2026	
	Diagnostic Status: Warning												
	Observing Template: MIRI Coronagraphic Imaging												
	Background Observations:[BKG-2MJ0227_PSF_reference (Obs 8)]												
Diagnostics	(Visit 7:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous			
	(4)	2MJ0227_psf_reference	RA: 02 02 18.8576 (30.5785733d) Dec: +37 21 39.39 (37.36094d) Equinox: J2000				Proper Motion RA: -4.974 mas/yr Proper Motion Dec: -6.6029999516104 mas/yr Parallax: 7.354000000000001E-4" Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.												
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.												
	Category=Star Description=[K dwarfs, K stars] Extended=NO												
Acquisition	#	Target	Filter	Quadrant	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID			
	1	4 2MJ0227_psf_reference	FND	1	FAST	4	1	1	0.959	276758.7			
Template	Repeat observation												
	NO												
Dithers	#	Dither Type											
	1	9-POINT-SMALL-GRID											
Spectral Elements	#	Coron Mask/Filter	Subarray	Mask	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	4QPM/F1140C	MASK1140	4QPM	F1140C	FASTR1	494	5	1	9	45	5336.715	276758.8

Proposal 11225 - Observation 7 - A MIRI search for TWA 7b twins: sub-Jupiter mass planets shaping debris disks

PSF References	PSF Reference: true
Special Requirements	No Parallel Attachments Sequence Observations 5, 6, 7, 8, Non-interruptible

Proposal 11225 - Observation 8 - A MIRI search for TWA 7b twins: sub-Jupiter mass planets shaping debris disks

Observation	Proposal 11225, Observation 8: BKG-2MJ0227_PSF_reference												Mon Jun 15 18:00:28 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Coronagraphic Imaging												
	Background Observation For: [2MJ0227_PSF_reference (Obs 7)]												
Diagnostics	(Visit 8:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates					Targ. Coord. Corrections			Miscellaneous		
	(12)	BKG-2MJ0227_psf_reference	RA: 02 02 56.0000 (30.7333333d) Dec: +37 23 35.00 (37.39306d) Equinox: J2000										
	Comments: Category=Calibration Description=[Coronagraphic, Telescope/sky background] Extended=NO												
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Repeat observation					Background Quadrant						
		YES					1						
Dithers	#	Dither Type											
	1	BACKGROUND											
Spectral Elements	#	Coron Mask/Filter	Subarray	Mask	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	4QPM/F1140C	MASK1140	4QPM	F1140C	FASTR1	494	5	1	2	10	1185.937	276758.8
PSF References	Additional Justification: false												

Proposal 11225 - Observation 8 - A MIRI search for TWA 7b twins: sub-Jupiter mass planets shaping debris disks

Special Requirements	No Parallel Attachments Sequence Observations 5, 6, 7, 8, Non-interruptible
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Proposal 11225 - Observation 9 - A MIRI search for TWA 7b twins: sub-Jupiter mass planets shaping debris disks

Observation	Proposal 11225, Observation 9: BKG_Fomalhaut_C Mon Jun 15 18:00:28 GMT 2026												
	Diagnostic Status: Warning Observing Template: MIRI Coronagraphic Imaging Background Observation For: [Fomalhaut_C (Obs 10)]												
Diagnostics	(Visit 9:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous			
	(13)	BKG-Fomalhaut_C	RA: 22 48 42.0000 (342.1750000d) Dec: -24 22 52.00 (-24.38111d) Equinox: J2000 <i>Comments:</i> <i>Category=Calibration</i> <i>Description=[Coronagraphic, Telescope/sky background]</i> <i>Extended=NO</i>										
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Repeat observation				Background Quadrant							
		YES				1							
Dithers	#	Dither Type											
	1	BACKGROUND											
Spectral Elements	#	Coron Mask/Filter	Subarray	Mask	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	4QPM/F1140C	MASK1140	4QPM	F1140C	FASTR1	1251	12	1	2	24	7201.425	276758.10
PSF References	Additional Justification: false												

Proposal 11225 - Observation 9 - A MIRI search for TWA 7b twins: sub-Jupiter mass planets shaping debris disks

Special Requirements	No Parallel Attachments Sequence Observations 9, 10, 11, 12, Non-interruptible
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Proposal 11225 - Observation 10 - A MIRI search for TWA 7b twins: sub-Jupiter mass planets shaping debris disks

Observation	Proposal 11225, Observation 10: Fomalhaut_C												Mon Jun 15 18:00:28 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Coronagraphic Imaging												
	Background Observations:[BKG_Fomalhaut_C (Obs 9)]												
Diagnostics	(Fomalhaut_C (Obs 10)) Warning (Form): Science observations should be linked to at least one other compatible science observation by an Aperture PA Offset of 1-14 degrees												
	(Visit 10:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous			
	(5)	Fomalhaut_C	RA: 22 48 4.4929 (342.0187204d) Dec: -24 22 7.72 (-24.36881d) Equinox: J2000				Proper Motion RA: 331.609 mas/yr Proper Motion Dec: -183.80500009698153 mas/yr Parallax: 0.130270700000000002" Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.												
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.												
	Category=Star Description=[Circumstellar dust, Circumstellar gas, Debris disks, M dwarfs] Extended=NO												
Acquisition	#	Target	Filter	Quadrant	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID			
	1	5 Fomalhaut_C	FND	1	FAST	4	1	1	0.959	276758.9			
Template	Repeat observation												
	NO												
Dithers	#	Dither Type											
	1	NONE											
Spectral Elements	#	Coron Mask/Filter	Subarray	Mask	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	4QPM/F1140C	MASK1140	4QPM	F1140C	FASTR1	1251	12	1	1	12	3600.713	276758.10

Proposal 11225 - Observation 10 - A MIRI search for TWA 7b twins: sub-Jupiter mass planets shaping debris disks

PSF References	Fomalhaut_C_PSF_reference (Obs 11) (PSF Reference; Filters [F1140C]) Additional Justification: false
Special Requirements	No Parallel Attachments Sequence Observations 9, 10, 11, 12, Non-interruptible

Proposal 11225 - Observation 11 - A MIRI search for TWA 7b twins: sub-Jupiter mass planets shaping debris disks

Observation	Proposal 11225, Observation 11: Fomalhaut_C_PSF_reference											Mon Jun 15 18:00:28 GMT 2026	
	Diagnostic Status: Warning												
	Observing Template: MIRI Coronagraphic Imaging												
	Background Observations:[BKG-Fomalhaut_PSF_reference (Obs 12)]												
Diagnostics	(Visit 11:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous			
	(6)	Fomalhaut_C_psf_reference	RA: 23 32 49.3982 (353.2058258d) Dec: -16 50 44.32 (-16.84564d) Equinox: J2000				Proper Motion RA: 341.147 mas/yr Proper Motion Dec: -219.10799994202534 mas/yr Parallax: 0.06873560000000001" Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.												
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.												
	Category=Star Description=[K dwarfs] Extended=NO												
Acquisition	#	Target	Filter	Quadrant	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID			
	1	6 Fomalhaut_C_psf_reference	FND	1	FAST	36	1	1	8.628	276758.11			
Template	Repeat observation												
	NO												
Dithers	#	Dither Type											
	1	9-POINT-SMALL-GRID											
Spectral Elements	#	Coron Mask/Filter	Subarray	Mask	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	4QPM/F1140C	MASK1140	4QPM	F1140C	FASTR1	708	5	1	9	45	7644.833	276758.12

Proposal 11225 - Observation 11 - A MIRI search for TWA 7b twins: sub-Jupiter mass planets shaping debris disks

PSF References	PSF Reference: true
Special Requirements	No Parallel Attachments Sequence Observations 9, 10, 11, 12, Non-interruptible

Proposal 11225 - Observation 12 - A MIRI search for TWA 7b twins: sub-Jupiter mass planets shaping debris disks

Observation	Proposal 11225, Observation 12: BKG-Fomalhaut_PSF_reference												Mon Jun 15 18:00:28 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Coronagraphic Imaging												
	Background Observation For: [Fomalhaut_C_PSF_reference (Obs 11)]												
Diagnostics	(Visit 12:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates					Targ. Coord. Corrections			Miscellaneous		
	(14)	BKG-Fomalhaut_C_psf_reference	RA: 23 33 13.0000 (353.3041667d) Dec: -16 57 5.00 (-16.95139d) Equinox: J2000										
	Comments: Category=Calibration Description=[Coronagraphic, Telescope/sky background] Extended=NO												
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Repeat observation					Background Quadrant						
		YES					1						
Dithers	#	Dither Type											
	1	BACKGROUND											
Spectral Elements	#	Coron Mask/Filter	Subarray	Mask	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	4QPM/F1140C	MASK1140	4QPM	F1140C	FASTR1	708	5	1	2	10	1698.852	276758.12
PSF References	Additional Justification: false												

Proposal 11225 - Observation 12 - A MIRI search for TWA 7b twins: sub-Jupiter mass planets shaping debris disks

Special Requirements	No Parallel Attachments Sequence Observations 9, 10, 11, 12, Non-interruptible
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Proposal 11225 - Observation 13 - A MIRI search for TWA 7b twins: sub-Jupiter mass planets shaping debris disks

Observation	Proposal 11225, Observation 13: BKG-AU_Mic Mon Jun 15 18:00:28 GMT 2026												
	Diagnostic Status: Warning Observing Template: MIRI Coronagraphic Imaging Background Observation For: [AU_Mic (Obs 14)]												
Diagnostics	(Visit 13:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous			
	(15)	BKG-AU_Mic	RA: 20 44 36.0000 (311.1500000d) Dec: -31 25 5.00 (-31.41806d) Equinox: J2000										
	Comments: Category=Calibration Description=[Coronagraphic, Telescope/sky background] Extended=NO												
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Repeat observation				Background Quadrant							
		YES				1							
Dithers	#	Dither Type											
	1	BACKGROUND											
Spectral Elements	#	Coron Mask/Filter	Subarray	Mask	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	4QPM/F1140C	MASK1140	4QPM	F1140C	FASTR1	1251	12	1	2	24	7201.425	276758.14
PSF References	Additional Justification: false												

Proposal 11225 - Observation 13 - A MIRI search for TWA 7b twins: sub-Jupiter mass planets shaping debris disks

Special Requirements	No Parallel Attachments
	Sequence Observations 13, 14, 15, 16, Non-interruptible

Proposal 11225 - Observation 14 - A MIRI search for TWA 7b twins: sub-Jupiter mass planets shaping debris disks

Observation	Proposal 11225, Observation 14: AU_Mic											Mon Jun 15 18:00:28 GMT 2026	
	Diagnostic Status: Warning												
	Observing Template: MIRI Coronagraphic Imaging												
	Background Observations:[BKG-AU_Mic (Obs 13)]												
Diagnostics	(AU_Mic (Obs 14)) Warning (Form): Science observations should be linked to at least one other compatible science observation by an Aperture PA Offset of 1-14 degrees												
	(Visit 14:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous			
	(7)	AU_Mic	RA: 20 45 9.5325 (311.2897188d)				Proper Motion RA: 281.319 mas/yr						
	Dec: -31 20 27.24 (-31.34090d)				Proper Motion Dec: -360.1480000497759 mas/yr								
	Equinox: J2000				Parallax: 0.1029432"								
					Epoch of Position: 2000								
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.												
SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.													
Category=Star													
Description=[Circumstellar disks, Circumstellar dust, Debris disks, M dwarfs]													
Extended=NO													
Acquisition	#	Target	Filter	Quadrant	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID			
	1	7 AU_Mic	FND	1	FAST	4	1	1	0.959	276758.13			
Template	Repeat observation												
	NO												
Dithers	#	Dither Type											
	1	NONE											
Spectral Elements	#	Coron Mask/Filter	Subarray	Mask	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	4QPM/F1140C	MASK1140	4QPM	F1140C	FASTR1	1251	12	1	1	12	3600.713	276758.14

Proposal 11225 - Observation 14 - A MIRI search for TWA 7b twins: sub-Jupiter mass planets shaping debris disks

PSF References	AU_Mic_PSF_reference (Obs 15) (PSF Reference; Filters [F1140C]) Additional Justification: false
Special Requirements	Aperture PA Range 78.5 to 98.5 Degrees (V3 73.66455103 to 93.66455103) No Parallel Attachments Sequence Observations 13, 14, 15, 16, Non-interruptible

Proposal 11225 - Observation 15 - A MIRI search for TWA 7b twins: sub-Jupiter mass planets shaping debris disks

Observation	Proposal 11225, Observation 15: AU_Mic_PSF_reference											Mon Jun 15 18:00:28 GMT 2026	
	Diagnostic Status: Warning												
	Observing Template: MIRI Coronagraphic Imaging												
	Background Observations:[BKG-AU_Mic_PSF_reference (Obs 16)]												
Diagnostics	(Visit 15:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous				
	(8)	AU_Mic_psf_reference	RA: 20 13 53.3962 (303.4724842d) Dec: -45 09 50.47 (-45.16402d) Equinox: J2000			Proper Motion RA: 778.331 mas/yr Proper Motion Dec: -159.93899990007776 mas/yr Parallax: 0.16221709999999998" Epoch of Position: 2000							
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.												
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.												
	Category=Star												
	Description=[M dwarfs, M stars]												
Acquisition	Extended=NO												
	#	Target	Filter	Quadrant	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID			
	1	8 AU_Mic_psf_refere nce	FND	1	FAST	4	1	1	0.959	276758.15			
Template	Repeat observation												
	NO												
Dithers	#	Dither Type											
	1	9-POINT-SMALL-GRID											
Spectral Elements	#	Coron Mask/Filter	Subarray	Mask	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	4QPM/F1140C	MASK1140	4QPM	F1140C	FASTR1	1251	8	1	9	72	21603.557	276758.16

Proposal 11225 - Observation 15 - A MIRI search for TWA 7b twins: sub-Jupiter mass planets shaping debris disks

PSF References	PSF Reference: true
Special Requirements	No Parallel Attachments Sequence Observations 13, 14, 15, 16, Non-interruptible

Proposal 11225 - Observation 16 - A MIRI search for TWA 7b twins: sub-Jupiter mass planets shaping debris disks

Observation	Proposal 11225, Observation 16: BKG-AU_Mic_PSF_reference												Mon Jun 15 18:00:28 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Coronagraphic Imaging												
	Background Observation For: [AU_Mic_PSF_reference (Obs 15)]												
Diagnostics	(Visit 16:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates					Targ. Coord. Corrections			Miscellaneous		
	(16)	BKG-AU_Mic_psf_reference	RA: 20 13 26.1000 (303.3587500d) Dec: -45 08 38.15 (-45.14393d) Equinox: J2000										
	Comments: Category=Calibration Description=[Coronagraphic, Telescope/sky background] Extended=NO												
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Repeat observation					Background Quadrant						
		YES					1						
Dithers	#	Dither Type											
	1	BACKGROUND											
Spectral Elements	#	Coron Mask/Filter	Subarray	Mask	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	4QPM/F1140C	MASK1140	4QPM	F1140C	FASTR1	1251	8	1	2	16	4800.79	276758.16
PSF References	Additional Justification: false												

Proposal 11225 - Observation 16 - A MIRI search for TWA 7b twins: sub-Jupiter mass planets shaping debris disks

Special Requirements	No Parallel Attachments
	Sequence Observations 13, 14, 15, 16, Non-interruptible